

IS492 Spring 2026
CheckPoint 3

Injury Aware Workout Adapter

A workout adapter and generator that helps users modify training plans and avoid secondary injuries

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GitHub repo link:

<https://github.com/IS492-SP26/team-project-injury-aware-workout-planner>

- **Final Problem and Target Users**
- **Live Demo / Walkthrough**
- **Design Decisions & Challenges**
- **Safety, Privacy & Reliability**
- **Next Steps**

**Our
Agenda**

Problem and Target Users



» Who is our target audience?

Regular gym users with non-severe injuries (e.g., knee or shoulder) and in the recovery phase, but not in formal rehab, and want to stay active safely.

» What problem are you solving?

Current fitness apps don't adapt to users with injuries. We address the lack of safe, personalized workout guidance for these users.

» How do you claim your tool is successful?

Our tool is successful if users can continue working out safely without worsening their injury. Specifically, if they feel more confident and spend less time searching for guidance.

User Data Collected



- Collect basic user details: name, age, gender
- Ask injury-related information: type and date of injury
- Assess pain level through guided actions (not direct input)
- Example: ask user to perform a squat or arm movement
- User reports pain level using a slider based on the action
- Standardized movements help ensure consistent pain measurement

Walkthrough-UI

AI Error Handling

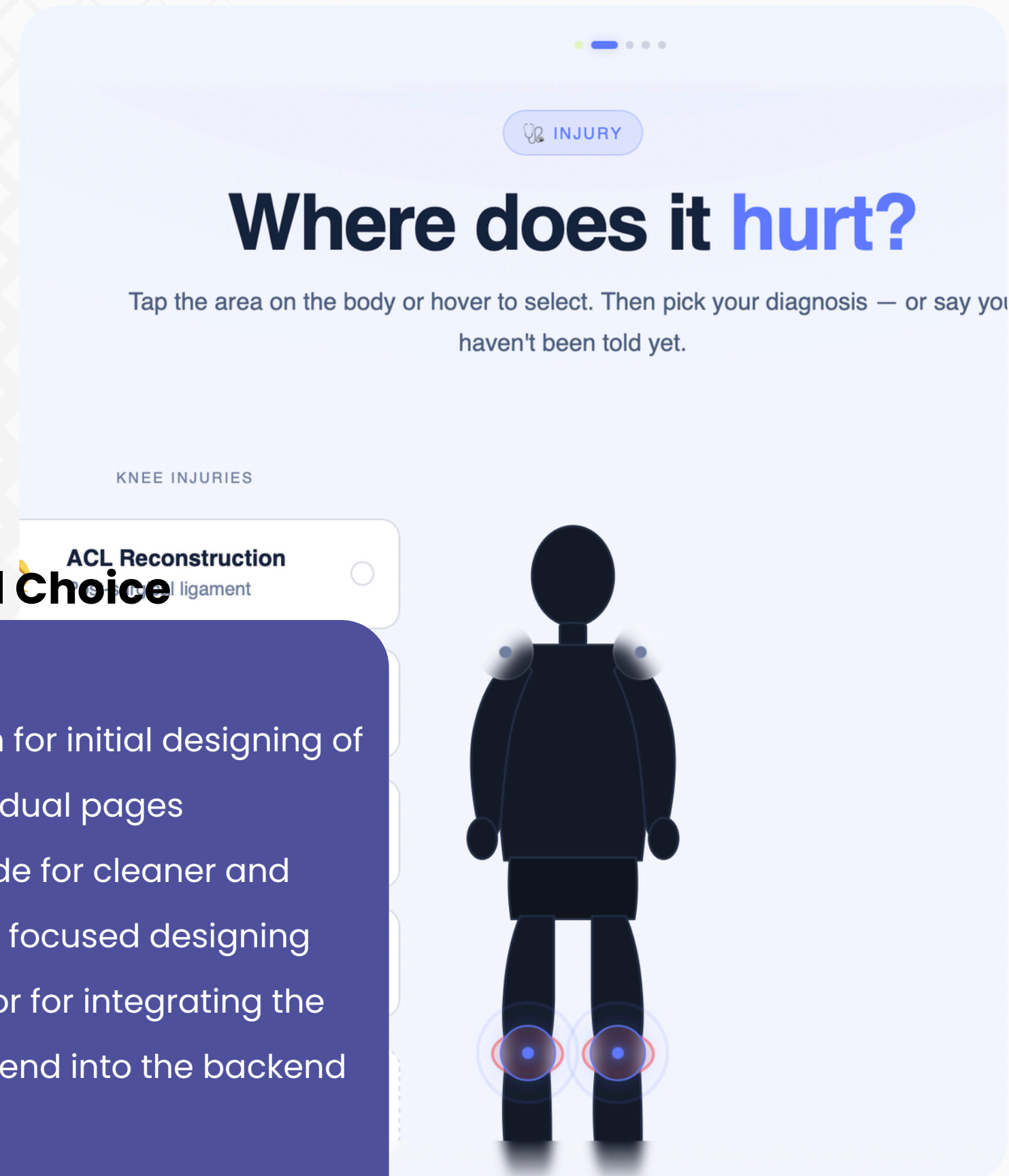
- AI hallucinations with incorrect placement and placing the cards in random places were handled by meticulously defining the placement instead of that back and forth prompting

Bug Fixes

- Adjusted the questionnaire to fit user needs, rather than keeping 20+ pages as seen on most recovery app
- Earlier had no interesting schema, too plain with no user functionality or didn't seem interesting

Model Choice

- Stitch for initial designing of individual pages
- Claude for cleaner and more focused designing
- Cursor for integrating the front end into the backend



Walkthrough-video process & groq API call

AI Error Handling

- Plan-first approach (Cursor Plan Mode) to improve output consistency
- Reduces randomness and aligns model behavior with expectations

Bug Fixes

- Fixed inconsistency between frontend input and backend data structure
- Ensures smooth and reliable data flow across the system

Model Choice

- Groq model for workout modification
- Fast, free, and suitable for real-time demo performance





Walkthrough Vercel Deployment

DEMO



AI Error Handling

- Replaced brittle file-based session dependence with request-safe payload handling
- Added clearer API error surfacing for survey save, login, and workout modification
- Added Gemini fallback behavior for temporary model overloads
- Ensured production config is loaded at runtime instead of hardcoding secrets

Bug Fixes

- Fixed yt-dlp dependency issues by replacing YouTube metadata extraction with Gemini-based structured prompting for timestamps
- Fixed Vercel session persistence issues by making workout modification independent of temporary server-side settings.

Model Choice

- Gemini is used for YouTube URL to timestamp convert.
- Gemini also verifies and corrects the timeline.
- Groq remains supported for workout adaptation locally, but production currently runs on Gemini-only config
- Added fallback Gemini model strategy to handle high-demand spikes.

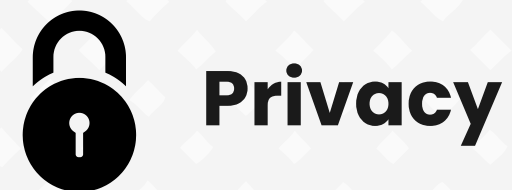


Safety, Privacy & Reliability

We prioritize safety, data privacy, and reliable AI behavior in our system design.



- Not intended as medical advice — for fitness guidance only
- Provide warnings for high-risk conditions
- Avoid unsafe or harmful workout recommendations



- Collect minimal user data (name, age, height, weight, injury-related inputs)
- Data is used only for generating recommendations
- Do not store sensitive personal or medical data

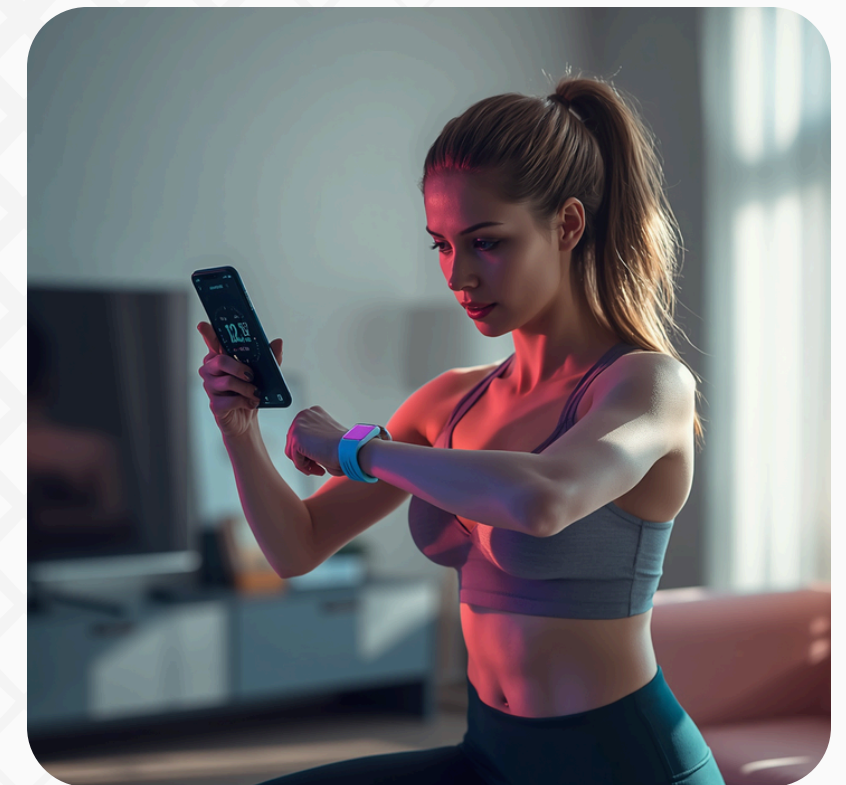


- Use structured inputs to reduce ambiguity
- Apply a plan-first approach to improve AI output consistency
- Keep API usage lightweight to avoid rate limit issues



Next Steps

- » Improve compatibility with diverse YouTube workout content, including videos without structured chapters
- » Expand the system to support more injuries and body areas beyond shoulder and knee



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Thank You •

